

CHEN-HUA Hsu 許宸華

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QUALIFICATIONS / SUMMARY

- AI/ML Engineer with an M.S. in Engineering Science focused on multimodal AI research and model optimization, with experience in VLM reliability, inference efficiency, systems integration, and edge AI.
- Hands-on experience in IT integration using Python backends, JavaScript interfaces, and APIs, complemented by AIoT integration of Raspberry Pi camera and sensor data, VLM/RAG analysis, and LINE Bot notifications.
- Achieved **1.9×** the inference throughput of Online VCD; in separate video-VLM research, reduced computation by **24%** while improving Top-1 accuracy by **6 percentage points**.

EDUCATION

National Cheng Kung University

Sep. 2024 – Jul. 2026

Master - Department of Engineering Science

Tainan, Taiwan

- Relevant Coursework: Deep Learning, Generative AI, Natural Language Processing, Optimization.

National Cheng Kung University

Sep. 2020 – Jul. 2024

Bachelor - Department of Engineering Science

Tainan, Taiwan

- Relevant Coursework: Machine Learning, Data Structures, Algorithms, Discrete Mathematics.

THESIS

Distilling Visual Contrastive Decoding into a Single Model for Enhanced Inference and Hallucination Mitigation

| Master's Thesis, Knowledge Distillation, Hallucination Mitigation, LoRA

Jul. 2026

- Proposed Visual Contrastive Decoding Distillation (VCDD) for LLaVA-1.5-7B, transferring token-level anti-hallucination signals into a LoRA adapter using Smooth-AW KL distillation and anti-forgetting regularization.
- Improved F1 over baseline across all 9 POPE settings and accuracy in **8 of 9**; increased MME score from **1560.12 to 1575.89** and LLaVA-Bench Composite score from **4.050** to a three-run mean of **4.219**.
- Reduced inference latency by **47.4%** versus Online VCD (**270 ms to 142 ms/query**), delivering **1.9×** the throughput of Online VCD over 3,000 POPE queries while restoring single-path inference.

PROJECTS

Video-Based Attention-Query Filtering Technique | Patent Application, Python, PyTorch, VLM

Jan. 2025 – Dec. 2025

- Co-developed foreground/background query filtering and pooling; implemented the prototype and ablation studies in Python/PyTorch to evaluate efficiency-accuracy trade-offs.
- Reduced GFLOPs by **24%** while improving Top-1 accuracy by **6 percentage points**, profiled on **768** videos with **16** frames each; Taiwan patent application **P52140051TW** pending.

Transformer Coil Winding AI Recommendation System | Full-Stack, Python, JavaScript, APIs

Oct. 2024 – Sep. 2025

- Developed a regression system mapping transformer specifications to winding parameters, reducing manual trial-and-error in engineering design workflows.
- Built a website for engineers to obtain real-time winding recommendations, connecting a JavaScript interface to the regression model through a Python API.

Intelligent Security System Integrating VLM and RAG | AIoT, RAG, Raspberry Pi, LINE Bot

Sep. 2024 – Dec. 2024

- Built a multimodal campus-safety pipeline combining VLM-based camera-scene understanding with temperature and humidity sensor data for risk assessment.
- Integrated Raspberry Pi edge collection, RAG grounded in safety guidelines and incident SOPs, and a LINE Bot for real-time alerts; documented evaluation criteria and key failure cases.

TECHNICAL SKILLS

AI / ML: PyTorch, Machine Learning, Deep Learning, VLMs, LLMs, LoRA, Knowledge Distillation, RAG

Optimization / Evaluation: Model Optimization, Model Evaluation, Inference Profiling

Programming / Integration: Python, JavaScript, C++, API Integration

Data Analysis: NumPy, pandas, Matplotlib

Tools / Platforms: Git, Docker, Linux (basic), Raspberry Pi